

Mod 1 Solution

Q.1) How Does IT Impact Organizations? / Explain the impact of information system on organization and society / Explain the importance of Information systems to Society.

Introduction

Information Technology (IT) and **Information Systems (IS)** have significantly changed how organizations and societies function. They provide tools for communication, data storage, decision-making, and automation, creating both opportunities and challenges.

A. Impact on Organizations:

- **Efficiency and Productivity:**
 - IT helps automate repetitive tasks, reduce manual errors, and speed up overall workflows across departments. This leads to lower operational costs and higher output.
 - *Example:* Enterprise Resource Planning (ERP) systems integrate finance, HR, supply chain, and procurement, allowing information to flow smoothly and reducing delays caused by disconnected processes.
- **Better Decision-Making:**
 - Real-time data from Management Information Systems (MIS), Business Intelligence (BI) dashboards, and analytics tools allows managers to make more precise strategic and tactical decisions. These insights help organizations respond quickly to market changes.
 - *Example:* Retailers analyze sales trends and customer behavior to forecast demand, optimize inventory levels, and reduce stockouts or overstock situations.

- **Improved Communication and Collaboration:**
 - Digital tools such as email, cloud-based platforms, instant messaging, and video conferencing enhance internal communication. They enable teams located in different regions to collaborate seamlessly and maintain continuous workflow.
 - *Example:* Remote teams can co-edit documents in real time using platforms like Google Workspace or Microsoft 365.
 - **Customer Engagement and Service:**
 - IT enables companies to interact with customers across multiple channels and deliver personalized services based on customer data.
 - *Example:* Multinational companies conduct global meetings through Zoom or MS Teams, ensuring smooth communication with clients and partners worldwide.
 - Customer Relationship Management (CRM) tools and AI-powered chatbots help businesses understand customer needs, offer tailored recommendations, and provide faster support.
 - *Example:* Banks use mobile apps and digital platforms to offer 24/7 services such as fund transfers, account monitoring, and loan applications.
 - **Competitive Advantage:**
 - Organizations adopt advanced technologies like Artificial Intelligence (AI), the Internet of Things (IoT), machine learning, and cloud computing to innovate faster and outperform competitors. Leveraging these technologies allows businesses to improve products, enhance operational efficiency, and respond rapidly to customer expectations.
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B. Impact on Society

1. Accessibility of Information:

- The internet, search engines, and digital libraries make vast amounts of knowledge easily accessible to people regardless of location or background. This supports continuous learning and reduces dependence on traditional sources.

- *Example:* Online education platforms like Coursera, SWAYAM, edX, and Khan Academy allow learners to access high-quality courses from universities and experts around the world.

2. Improved Services:

- IT enhances essential services by making them faster, more accurate, and more convenient. In healthcare, telemedicine helps patients consult doctors remotely; in governance, digital platforms reduce paperwork and improve transparency; in finance, technologies like UPI, digital wallets, and online banking enable secure, real-time transactions.
- These improvements save time, reduce costs, and increase accessibility for people in both urban and rural areas.

3. Global Connectivity:

- Social media, messaging platforms, and video-calling apps make it possible for individuals, businesses, and communities to stay connected across continents. This fosters cultural exchange, remote collaboration, and global awareness.
- These technologies also support online communities where people share ideas, work, hobbies, and social causes.

4. Challenges:

- Despite its benefits, IT also brings several challenges. Cybersecurity threats and data breaches put personal and organizational information at risk. Privacy concerns arise as more data is collected online. Automation and AI can lead to job displacement in certain sectors.
- Additionally, the digital divide—differences in access to devices, connectivity, and digital literacy—means not everyone benefits equally from technological progress.

Q.2) Explain in detail computer-based information systems(CBIS) and list various components of it.

Introduction

A **Computer-Based Information System (CBIS)** is a system that uses computer technology to collect, process, store, and distribute information to support decision-making in organizations. It includes both **types** of systems and their **components**.

Unlike manual systems, a CBIS relies on computer hardware and software to perform tasks efficiently, ensuring that managers and employees have timely access to accurate information.

Types of CBIS:

1. Transaction Processing System (TPS):

- Handles routine, repetitive business transactions such as sales, payroll, billing, order processing, and inventory updates. These systems ensure accuracy, speed, and consistency in daily operations.
- They form the foundation for higher-level systems because the data they collect is used for reporting and analysis.

2. Management Information System (MIS):

- Provides periodic summarized reports to middle managers to support planning, monitoring, and decision-making. These reports are usually structured and based on data generated by TPS.
- MIS helps managers identify trends, compare performance against goals, and ensure smooth functioning of departmental activities.

3. Decision Support System (DSS):

- Assists managers in making non-routine or complex decisions by offering interactive tools, analytical models, and simulations.
- DSS is particularly useful in situations with uncertainty, allowing users to test different scenarios and evaluate possible outcomes before choosing the best option.

4. Executive Support System (ESS):

- Provides top-level executives with strategic information that supports long-term planning and organizational direction.
- These systems often include dashboards, trend analyses, forecasting tools, and external data, helping executives monitor overall

performance and spot emerging opportunities or threats.

5. Office Automation System (OAS):

- Supports everyday office tasks such as document creation, data storage, communication, and scheduling. Tools like word processors, spreadsheets, email systems, and calendar applications fall under this category.
- OAS increases individual productivity and enhances team collaboration by simplifying routine administrative work.

6. Knowledge Management System (KMS):

- Helps organizations capture, store, organize, and share knowledge, expertise, and best practices. This includes documents, project experiences, manuals, and employee insights.
- KMS ensures that valuable knowledge is retained even when employees leave and promotes continuous learning within the organization.

7. Expert System (ES):

- Uses Artificial Intelligence techniques to mimic the decision-making ability of a human expert. These systems rely on knowledge bases and inference engines to provide specialized solutions.
- *Example:* Medical diagnosis systems, technical troubleshooting tools, or credit-risk evaluation systems that offer expert-level recommendations.

Components of CBIS

A CBIS consists of six critical components that must work together to produce value.³ These are often referred to as the resources of an information system.

1. Hardware (The Physical Layer)

Hardware refers to the tangible physical equipment used to perform input, processing, and output activities.⁴

- **Key Elements:** It includes computers (servers, workstations), input devices (keyboards, scanners), output devices (printers, monitors), and storage devices (hard drives).⁵
- **Role:** It provides the computing power required to run software and store data.

2. Software (The Instruction Layer)

Software consists of the programs and instructions that tell the hardware what to do.⁶ It bridges the gap between the user and the hardware.

- **System Software:** Manages the hardware (e.g., Windows, Linux, Android).⁷
- **Application Software:** Performs specific business tasks (e.g., Microsoft Excel for calculations, SAP for ERP, Salesforce for CRM).⁸

3. Data (The Raw Material)

Data refers to raw facts, figures, and statistics that have not yet been processed.⁹

- **Example:** Employee names, inventory levels, or daily sales figures.
- **Role:** In a CBIS, data is stored in **databases** where it can be organized and retrieved easily.¹⁰ Once processed, this data becomes useful *information* for decision-making.

4. Procedures (The Rules)

Procedures are the policies, methods, and rules that govern the operation of the computer system.¹¹

- **Role:** They define how the system should be used, who has access to it, and how to handle emergencies (like data backups or system failures). Procedures ensure the system is used consistently and securely.

5. People (The Human Element)

People are the most important component because they build, manage, and use the system.¹²

- **End-Users:** The individuals who use the system to do their daily jobs (e.g., accountants, clerks, managers).¹³
- **IS Specialists:** The technical staff who develop and maintain the system (e.g., programmers, system analysts, network administrators).¹⁴

6. Network / Telecommunications

This component connects the hardware together to form a network, allowing different parts of the system to communicate.¹⁵

- **Role:** It enables resource sharing (like printers) and communication (like email and internet access), allowing data to flow between different departments or geographic locations.

Key Takeaways for Revision

- **CBIS:** Integration of tech and people to process data.¹⁶
 - **6 Components:** Hardware, Software, Data, Procedures, People, Network.
 - **Distinction:** Hardware is the machine; Software is the instruction; Procedures are the rules for usage.
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Q.3) What are types IS? Explain with example / What are the different types of MIS?

Introduction

Organizations are divided into different levels (Operational, Tactical, and Strategic), and each level requires a different type of Information System to support its specific decision-making needs. This hierarchy is often represented as the "Pyramid of Information Systems." Getty Images

There are **four major types** of Information Systems used in organizations:

1. Transaction Processing Systems (TPS)

- **Level:** Operational Level (Bottom of the pyramid).
- **Users:** Operational staff, supervisors, clerks.
- **Definition:** TPS are computerized systems that perform and record the daily routine transactions necessary to conduct business. They bridge the gap between the organization and its external environment (customers, vendors).
- **Key Characteristics:**
 - **High Volume:** Processes thousands of transactions daily.
 - **Structured:** Follows predefined rules (e.g., if Credit > Limit, then Reject).
 - **Real-Time:** Data is updated instantly.
- **Examples:**

- **ATM Systems:** Records cash withdrawals and updates balances instantly.
- **Payroll Systems:** Calculates wages based on hours worked.
- **Point of Sale (POS):** Scans items at a supermarket checkout.

2. Management Information Systems (MIS)

- **Level:** Management / Tactical Level (Middle of the pyramid).
- **Users:** Middle Managers.
- **Definition:** MIS provides reports on the organization's current performance. It takes the raw data from the **TPS** and converts it into summarized reports to help managers monitor and control the business.
- **Key Characteristics:**
 - **Routine:** Produces weekly, monthly, or quarterly reports.
 - **Summary-Oriented:** Focuses on aggregated data rather than individual details.
 - **Internal Focus:** Primarily looks at data generated inside the company.
- **Examples:**
 - **Sales Management Systems:** A weekly report showing total sales by region.
 - **Inventory Control Systems:** A monthly report showing items that are low in stock.

3. Decision Support Systems (DSS)

- **Level:** Management / Tactical Level (Middle/Upper pyramid).
- **Users:** Professionals, Senior Managers, Analysts.
- **Definition:** DSS are interactive systems designed to help managers make decisions in unique, rapidly changing, and non-routine situations (semi-structured problems). Unlike MIS, which tells you *what happened*, DSS helps you ask "*What if?*"
- **Key Characteristics:**
 - **Analytical:** Uses complex mathematical models and statistical tools.

- **Interactive:** The user can input different variables to see different outcomes.
- **Examples:**
 - **Loan Authorization System:** A bank manager inputs a client's credit score and income to decide whether to grant a loan.
 - **Voyage Estimating Systems:** A shipping company calculates the most profitable route for a ship based on fuel costs and cargo rates.

4. Executive Support Systems (ESS)

- **Level:** Strategic Level (Top of the pyramid).
- **Users:** Senior Executives (CEO, CFO, CTO).
- **Definition:** ESS (also called Executive Information Systems) helps senior leaders make long-term strategic decisions. These decisions are often unstructured and require high-level insights into trends, the economy, and competitors.
- **Key Characteristics:**
 - **Visual:** Uses "Dashboards" and graphs for a quick overview.
 - **External Data:** Incorporates external info like stock market trends, laws, and competitor news.
 - **Drill-Down:** Allows executives to see a summary and click deeper for details if needed.
- **Examples:**
 - **5-Year Operating Plan:** Predicting long-term market growth.
 - **Digital Dashboard:** A single screen showing the company's stock price, global revenue, and major competitor news live.

Key Takeaways for Revision

- **TPS: Doers.** Records daily data (ATM, Payroll).
- **MIS: Watchers.** Summarizes data for control (Sales Report).
- **DSS: Analyzers.** Helps solve "What if" problems (Loan Analysis).
- **ESS: Leaders.** Strategic planning with dashboards (5-Year Plan).

Q.4) Create MIS system for any hospital.

Introduction

An MIS for a hospital, often called a Hospital Management System (HMS) or Hospital Information System (HIS), is an integrated software solution designed to manage all aspects of hospital operations. It connects clinical, administrative, and financial departments into a single network to improve the quality of healthcare, reduce operational costs, and streamline the flow of information.

Key Functional Modules of Hospital MIS

To "create" this system, we must design specific subsystems (modules) that handle different functions. These modules share a central database so data enters once and is available everywhere.

1. Patient Administration Module (The Front Desk)

- **Registration:** Captures patient demographics (Name, Age, Contact) and generates a Unique Patient ID (UHID).
- **Scheduling:** Manages doctor appointments to prevent overcrowding.
- **ADT (Admission, Discharge, Transfer):** Tracks live bed availability. It records when a patient is admitted to a ward, moved to ICU, or discharged.

2. Clinical Modules (The Core Medical Service)

- **Electronic Medical Records (EMR):** Stores patient history, allergies, past surgeries, and current diagnosis. This eliminates paper files.
- **Doctor/Nursing Station:** Allows doctors to enter prescriptions and nurses to update vitals (BP, Temperature) directly into the system.
- **Operation Theatre (OT) Management:** Schedules surgeries and manages the inventory required for specific procedures.

3. Ancillary & Support Services

- **Laboratory Information System (LIS):** When a doctor orders a blood test, the request goes here instantly. Lab technicians input results, which are automatically linked to the patient's EMR.
- **Pharmacy Management:** Links to the doctor's prescription. It checks stock levels, dispenses medicine, and alerts if a medicine is nearing its expiry date.

- **Radiology (RIS):** Manages X-rays, CT scans, and MRIs.

4. Administrative & Financial Modules

- **Billing & Insurance:** Automatically collates charges from Room Rent + Doctor Fees + Lab Tests + Pharmacy to generate the final bill. It also handles insurance claims.
 - **HR & Payroll:** Manages staff shifts (rosters) and salaries.
 - **Inventory Control:** Tracks consumption of medical supplies (syringes, gloves) to prevent stockouts.
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Workflow Example (How the System Works)

1. **Input:** A patient arrives. The **Registration Module** generates a Patient ID.
 2. **Process:** The doctor views the ID in the **EMR Module**, diagnoses the patient, and orders a digital prescription.
 3. **Integration:** The **Pharmacy Module** sees the prescription instantly and packs the meds. Simultaneously, the **Billing Module** adds the consultation and medicine charges to the patient's account.
 4. **Output:** The patient pays the bill and receives a printed summary. Management gets a report on "Daily Revenue" and "Patient Footfall."
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Key Takeaways for Revision

- **Central Database:** The heart of HMS; connects all departments.
- **UHID:** Unique Patient ID links all records.
- **Modules:** Registration, EMR, Pharmacy, Labs, Billing.
- **Goal:** To move from "Paper-based" to "Paperless" efficiency.